ANNEX I

SUMMARY OF PRODUCT CHARACTERISTICS

1. NAME OF THE MEDICINAL PRODUCT

Ervebo solution for injection
Ebola Zaire Vaccine (rVSVΔG-ZEBOV-GP, live)

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

One dose (1 mL) contains:

Ebola Zaire Vaccine (rVSVΔG-ZEBOV-GP[1,2] live, attenuated) ≥ 72 million pfu[3]
[1]Recombinant Vesicular Stomatitis Virus (rVSV) strain Indiana with a deletion of the VSV envelope glycoprotein (G) replaced with the Zaire Ebola Virus (ZEBOV) Kikwit 1995 strain surface glycoprotein (GP)
[2]Produced in Vero cells
[3]pfu=plaque-forming units

This product contains genetically modified organisms (GMOs).
This vaccine contains a trace amount of rice protein. See section 4.3.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Solution for injection.
The solution is a colourless to slightly brownish-yellow liquid.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Ervebo is indicated for active immunisation of individuals 1 year of age or older to protect against Ebola Virus Disease (EVD) caused by Zaire Ebola virus (see sections 4.2, 4.4 and 5.1).

The use of Ervebo should be in accordance with official recommendations.

4.2 Posology and method of administration

Ervebo should be administered by a trained healthcare worker.

<u>Posology</u>

Individuals 1 year of age or older: one dose (1 mL) (see section 5.1).

Booster dose
The need and appropriate timing for booster dose(s) have not been established. Current available data are included in section 5.1.

Paediatric population

The posology in children 1 to 17 years of age is the same as in adults (see sections 4.8 and 5.1).

Ervebo should not be used in children aged less than 1 year of age because the safety, immunogenicity and efficacy have not been established (see sections 4.8 and 5.1).

Method of administration

For precautions to be taken before administering the vaccine, see section 4.4.

For precautions regarding thawing, handling and disposal of the vaccine, see section 6.6.

Ervebo should be administered by the intramuscular route. The preferred site is the deltoid area of the non-dominant arm or in the higher anterolateral area of the thigh. The vaccine should not be injected intravascularly. No data are available for administration via the subcutaneous or intradermal routes.

The vaccination injection site or any vesicles must be covered with an adequate bandage (e.g., any adhesive bandage or gauze and tape). This provides a physical barrier to protect against direct contact (see sections 4.4 and 5.3). The bandage may be removed when there is no visible fluid leakage.

The vaccine should not be mixed in the same syringe with any other vaccines or medicinal products.

4.3 Contraindications

Hypersensitivity to the active substances, to any of the excipients listed in section 6.1 or to rice protein listed in section 2.

4.4 Special warnings and precautions for use

Traceability

In order to improve the traceability of biological medicinal products, the name and the batch number of the administered product should be clearly recorded.

Hypersensitivity and anaphylaxis

It is recommended to closely monitor vaccinees following vaccination for the early signs of anaphylaxis or anaphylactoid reactions. As with all injectable vaccines, appropriate medical treatment and supervision should always be readily available in case of an anaphylactic event following the administration of the vaccine.

Duration of protection

Vaccination with Ervebo may not result in protection in all vaccinees. Vaccine efficacy in adults has been established in the period ≥10 to ≤31 days after vaccination, however the duration of protection is not known (see section 5.1). **The use of other Ebola control measures should therefore not be interrupted**.

Vaccination of contacts of Ebola cases should occur as soon as possible (see section 5.1).

Standard precautions when caring for patients with known or suspected Ebola disease

Vaccination with Ervebo does not eliminate the necessity of standard precautions when caring for patients with known or suspected Ebola disease. **All healthcare workers and other ancillary providers who have been vaccinated should not alter their practices with regard to safe injection, hygiene, and personal protective equipment (PPE) after vaccination.**

Healthcare workers caring for patients with suspected or confirmed Ebola virus should apply extra infection control measures to prevent contact with the patient's blood and body fluids and contaminated surfaces or materials such as clothing and bedding. Samples taken from humans and animals for investigation of Ebola infection should be handled by trained staff and processed in suitably equipped laboratories.

Vaccine administrators should counsel vaccinees to continue to protect themselves with adequate measures.

Immunocompromised individuals

Except for people living with HIV with infection controlled through antiretroviral therapy for whom safety and immunogenicity data are available (see sections 4.8 and 5.1), there are no data in immunocompromised individuals. Immunocompromised individuals may not respond as well as immunocompetent individuals to Ervebo. As a precautionary measure, it is preferable to avoid the use of Ervebo in individuals with known immunocompromised conditions or receiving immunosuppressive therapy, including the following conditions:

- Severe humoral or cellular (primary or acquired) immunodeficiency, e.g. severe combined immunodeficiency, agammaglobulinemia, or HIV infection with an AIDS defining illness or a CD4+ T-lymphocyte count below 200 cells/mm^3.
- Current immunosuppressive therapy, including high doses of corticosteroids. This does not include individuals who are receiving topical, inhaled or low-dose parenteral corticosteroids (e.g. for asthma prophylaxis or replacement therapy).
- Diseases of the blood such as leukaemia, lymphomas of any type, or other malignant neoplasms affecting the haematopoietic and lymphatic systems.
- Family history of congenital or hereditary immunodeficiency, unless the immune competence of the potential vaccine recipient is demonstrated.

Pregnant and breast-feeding women

As a precautionary measure, it is preferable to avoid the use of Ervebo during pregnancy. See section 4.6.

Transmission

Vaccine virus might be present in biological fluids such as blood, urine, saliva, semen, vaginal fluids, aqueous humor, breast milk, faeces, sweat, amniotic fluid, and placenta. In clinical studies, vaccine virus RNA has been detected by PCR in the plasma of most of the adult participants. Vaccine virus RNA was mainly detected from Day 1 to Day 7. Shedding of vaccine virus has been detected by PCR in urine or saliva in 19 out of 299 healthy adult participants and in skin vesicles in 4 out of 10 adult participants. The vaccine virus RNA was detected in a skin vesicle at 12 days post-vaccination in one of the four participants.

In a Phase 1 study, vaccine viremia and viral shedding were observed more frequently (28/39) in children and adolescents 6 to 17 years of age compared to adults. In a Phase 2 study, 31.7% (19/60) of children and adolescents 1 to 17 years of age enrolled in a shedding sub-study shed vaccine virus in saliva following vaccination. Viral shedding was observed more frequently on Day 7 and declined thereafter, with no shedding detected at Day 56. In a subsequent Phase 2 study of adolescents and adults living with HIV, 22.9% (46/201) had detectable vaccine virus RNA in saliva following vaccination. Viral shedding was observed more frequently on Day 7 and declined thereafter, with no shedding detected at Day 28.

Transmission of vaccine virus through close personal contact is accepted as a theoretical possibility. Vaccine recipients should avoid close contact with and exposure of high-risk individuals to blood and bodily fluids for at least 6 weeks following vaccination. High-risk individuals include:

- Immunocompromised individuals and individuals receiving immunosuppressive therapy (see section above),
- Pregnant or breast-feeding women (see section 4.6),

- Children < 1 year of age.

Individuals who develop vesicular rash after receiving the vaccine should cover the vesicles until they heal to minimise the risk of possible transmission of vaccine virus through open vesicles. Contaminated bandages must be disposed following institutional guidelines or WHO healthcare waste management policy. See section 5.3.

Parents and caregivers of young vaccinees should observe careful hygiene especially when handling bodily waste and fluids for a minimum of 6 weeks after vaccination. Disposable nappies can be sealed in double plastic bags and disposed of in household waste. See section 5.3.

Inadvertent transmission of vaccine virus to animals and livestock is also theoretically possible, see below.

Individuals administered Ervebo should not donate blood for at least 6 weeks post-vaccination.

Transmission to animals and livestock

Transmission of vaccine virus through close contact with livestock is accepted as a theoretical possibility. Vaccine recipients should attempt to avoid exposure of livestock to blood and bodily fluids for at least 6 weeks following vaccination. Individuals who develop vesicular rash after receiving the vaccine should cover the vesicles until they heal. Contaminated bandages must be disposed following institutional guidelines or WHO healthcare waste management policy. See section 5.3.

Concurrent illness

Vaccination should be postponed in individuals experiencing moderate or severe febrile illness. The presence of a minor infection should not result in deferral of vaccination.

Thrombocytopenia and coagulation disorders

The vaccine should be given with caution to individuals with thrombocytopenia or any coagulation disorder because bleeding or bruising may occur following an intramuscular administration in these individuals.

Protection against filovirus disease

The vaccine will not prevent disease caused by filoviruses other than Zaire Ebola virus.

Impact to serological testing

Following vaccination with Ervebo, individuals may test positive for Ebola glycoprotein (GP) nucleic acids, antigens, or antibodies against Ebola GP, which are targets for certain Ebola diagnostic tests. Therefore, diagnostic testing for Ebola should target non-GP sections of the Ebola virus.

Sodium

This medicinal product contains less than 1 mmol sodium (23 mg) per dose, that is to say essentially “sodium-free”.

4.5 Interaction with other medicinal products and other forms of interaction

No interaction studies have been performed.

As there are no data on co-administration of Ervebo with other vaccines, the concomitant use of Ervebo with other vaccines is not recommended.

Immune globulin (IG), blood or plasma transfusions should not be given concomitantly with Ervebo. Administration of immune globulins, blood or plasma transfusions administered 3 months before or up to 1 month after Ervebo administration may interfere with the expected immune response.

It is unknown whether concurrent administration of antiviral medication including interferons could impact vaccine virus replication and efficacy.

4.6 Fertility, pregnancy and lactation

Pregnancy

There are limited amount of data (less than 300 pregnancy outcomes) from the use of Ervebo in pregnant women, or women who became pregnant after receiving the vaccine. The safety of Ervebo has not been established in pregnant women.

As there are limitations to available data, including the small number of cases, caution should be exercised in drawing conclusions. Lack of reliable data on background rates of pregnancy and neonatal outcomes in the affected regions also makes a contextual assessment of the data challenging.

Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity (see section 5.3).

As a precautionary measure, it is preferable to avoid the use of Ervebo during pregnancy. Nevertheless, considering the severity of EVD, vaccination should not be withheld when there is a clear risk of exposure to Ebola infection.

Pregnancy should be avoided for 2 months following vaccination. Women of child-bearing potential should use an effective contraceptive method.

Breast-feeding

It is unknown whether the vaccine virus is secreted in human milk.

A risk to the newborns/infants from breast-feeding by vaccinated mothers cannot be excluded.

Evaluation of the vaccine virus in animal milk has not been conducted. When Ervebo is administered to female rats, antibodies against the vaccine virus were detected in offspring, likely due to acquisition of maternal antibodies via placental transfer during gestation and via lactation. See section 5.3.

A decision must be made whether to discontinue breast-feeding or to abstain from Ervebo taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman. In certain circumstances, where alternatives to breast-feeding are limited, the immediate need and health benefits to the infant should be taken into consideration and balanced with the mother’s need for Ervebo. Both may present compelling needs that should be considered before vaccination of the mother.

Fertility

There are no data on fertility effects in humans.

Animal studies in female rats do not indicate harmful effects (see section 5.3).

4.7 Effects on ability to drive and use machines

Ervebo has no or negligible influence on the ability to drive and use machines. However, some of the effects mentioned under section 4.8 “Undesirable effects” may temporarily affect the ability to drive or use machines.

4.8 Undesirable effects

Summary of the safety profile

For all age groups, anaphylaxis was reported very rarely (<1/10 000) in clinical studies.

In adults 18 years of age and older, the most common injection-site adverse reactions reported following vaccination with Ervebo were injection-site pain (70.3%), injection-site swelling (16.7%) and injection-site erythema (13.7%). The most common systemic adverse reactions were headache (55.1%), pyrexia (39.2%), myalgia (32.5%), somnolence, reduced activity, fatigue (25.5%), arthralgia (18.6%), chills (16.7%), decreased appetite (15.2%), abdominal pain (13.0%), nausea (9.5%), arthritis (3.7%), rash (3.6%), hyperhidrosis (3.2%), and mouth ulceration (2.2%). In general, these reactions were reported within 7 days after vaccination, were mild to moderate in intensity, and had short duration (less than 1 week).

In children and adolescents 1 to 17 years of age, the most common injection-site adverse reactions reported following vaccination with Ervebo were injection-site pain (41.6%), injection-site pruritus (4.1%), injection-site swelling (3.0%) and injection-site erythema (0.5%). The most common systemic adverse reactions were pyrexia (62.2%), headache (45.7%), somnolence, reduced activity, fatigue (23.5%), decreased appetite (23.4%), myalgia (15.8%), dizziness (9.9%), crying (6.4%) and mouth ulceration (2.5%). In general, these reactions were reported within 7 days after vaccination and were mild to moderate in intensity.

Tabulated list of adverse reactions

Frequencies are reported as:
Very common (≥1/10),
Common (≥1/100 to <1/10),
Uncommon (≥1/1 000 to <1/100),
Rare (≥1/10 000 to <1/1 000),
Very rare (<1/10 000),
Not known (cannot be estimated from the available data).
Within each frequency grouping, adverse reactions are presented in the order of decreasing seriousness.

Individuals 1 year of age and older
Table 1 shows the adverse reactions observed in recipients of Ervebo.

For adults, the frequencies listed are based on the higher frequency reported in the Phase 2/3 placebo-controlled randomised studies, Protocol 009, Protocol 012 and Protocol 016, that have included a total of 2 143 individuals.

For children and adolescents, the frequencies listed corresponds to those observed in Protocol 016, a Phase 2 placebo-controlled randomised study, that has included a total of 609 individuals (including 95 children from 1 to 3 years old, 310 children from 3 to 11 years old, and 204 children from 12 to 17 years old).

Table 1: Tabulated summary of adverse reactions in individuals 1 year of age and older

MedDRA-system organ class	Adverse reactions	Frequency	
		Children and adolescents¶	Adults*
Immune system disorders	Anaphylactic reaction	Very rare	Very rare
Nervous system disorders	Headache	Very common	Very common
	Dizziness	Common	Common
Gastrointestinal disorders	Abdominal pain	Very common	Very common
	Decreased appetite	Very common	Very common
	Nausea	Common	Common
Skin and subcutaneous tissue disorders	Mouth ulceration	Common	Common
	Rash§	None	Common
Musculoskeletal and connective tissue disorders	Arthralgia§	Common	Very common
	Myalgia	Very common	Very common
	Arthritis§	NA	Common
General disorders and administration site conditions	Pyrexia	Very common	Very common
	Somnolence†	Very common	Very common
	Chills	Very common	Very common
	Crying	Common	NA‡
	Injection site pain	Very common	Very common
	Injection site erythema	Uncommon	Very common
	Injection site pruritus	Common	Common
	Injection site swelling	Common	Very common
	Hyperhidrosis (sweats)	Common	Common

§See description of selected adverse reactions.
†Includes: somnolence, reduced activity and fatigue.
‡NA (not applicable): not assessed for this population.
¶The adverse reactions of abdominal pain, nausea, rash, arthralgia, chills, and hyperhidrosis occurred with a difference of <5% between vaccine and placebo groups.
*The adverse reactions of dizziness and injection site pruritus occurred with a difference of <5% between vaccine and placebo groups.

In a clinical study (Protocol 016), pyrexia was reported more frequently in younger children 1 to <3 years of age (83.2%), compared to children 3 to <12 years of age (64.8%), adolescents 12 to 17 years of age (48.3%) and adults (39.2%). Otherwise, the safety profile of Ervebo in children and adolescents 1 to 17 years of age was generally similar to that observed in adults.

Safety in adults and adolescents living with HIV

In Protocol 015, the safety profile of Ervebo in 52 adolescents and 149 adults living with HIV on antiretroviral therapy with an undetectable viral load and a CD4+ T-lymphocyte count greater than or equal to 200 cells/mm^3 was generally consistent with the known safety profile of Ervebo.

Description of selected adverse reactions

Arthralgia and arthritis

Arthralgia was generally reported in the first few days following vaccination, was mild to moderate in intensity, and resolved within one week after onset. Arthritis (arthritis, joint effusion, joint swelling, osteoarthritis, monoarthritis or polyarthritis) was generally reported within the first few weeks following vaccination. In clinical studies with reports of arthritis, the median onsets were between 10 and 12 days (range from 0 to 25 days). Arthritis has been reported by participants in clinical studies at a frequency that ranged from 0% in several protocols to 23.5% in one Phase 1 study. The majority of arthritis reactions were mild to moderate in severity. The median duration of arthritis across clinical studies in which arthritis was reported ranged from 2 days to 81.5 days (including duration of recurrent arthritis) with a maximum of 330 days. The reasons for differences in arthritis reporting across studies are not known but may be due to differences in study populations or outcome reporting. In the Phase 1 study with the highest rate of arthritis, 6 of 24 patients (25%) who reported arthritis after vaccination had persistent joint symptoms two years after vaccination. In a small number of participants, the vaccine virus was recovered from joint effusion samples, suggestive of a virally mediated process post-vaccination.

Rash
Rash was characterised in a variety of ways including generalised rash (2.3%), vesicular rash (0.5%), dermatitis (0.3%), or cutaneous vasculitis (0.01%) in clinical studies. In different studies, rash was reported with median onsets of 7.5 to 10.5 days (range from 0 to 47 days). The median durations reported were between 6 to 18 days. In 6 out of 18 participants tested, the vaccine virus was detected in rashes (described as dermatitis, vesicles or cutaneous vasculitis lesions) suggesting a virally mediated process post-vaccination.

Transient decrease in white blood cells
Transient decreases in counts of lymphocytes, neutrophils and total white blood cells in the first 3 days following vaccination have been observed very commonly in Phase 1/2 studies; these adverse reactions generally resolved after the first week post-vaccination. No adverse reactions of infections were observed in Phase 1/2 studies.

Reporting of suspected adverse reactions
Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system listed in Appendix V.

4.9 Overdose

No cases of overdose have been reported. In the event of overdose, monitoring of vital functions and possible symptomatic treatment is recommended.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Vaccines, Viral Vaccine, ATC code: J07BX02

Mechanism of action

Ervebo consists of a live, attenuated recombinant vesicular stomatitis virus-based vector expressing the envelope glycoprotein gene of Zaire Ebola virus (rVSVΔG-ZEBOV-GP). Immunisation with the vaccine results in an immune response and protection from Zaire Ebola Virus Disease (EVD). The relative contributions of innate, humoral and cell-mediated immunity to protection from Zaire Ebola virus are unknown.

Clinical immunogenicity and efficacy

The clinical development program included seven Phase 2/3 clinical studies (Protocols 009, 010, 011, 012, 015, 016 and 018). All participants received a single dose of vaccine except for a subset of participants in Protocol 002 (n=30), Protocol 015 (n=40) and Protocol 016 (n=385) who received two doses.

Clinical efficacy
Clinical efficacy of Ervebo in adults was assessed in Protocol 010.

Protocol 010 (Ring vaccination study) was a Phase 3 open-label cluster-randomised study of ring vaccination (vaccinating contacts and contacts of contacts [CCCs] of index Ebola cases) which evaluated efficacy and safety of Ervebo in Guinea. In this study, 9 096 participants ≥18 years of age who were considered CCCs of an index case with laboratory-confirmed EVD were randomised to immediate (4 539 participants in 51 clusters) or 21 days delayed (4 557 participants in 47 clusters) vaccination with Ervebo. Of those 9 096 participants, 4 160 received Ervebo (2 119 participants were vaccinated in the immediate arm and 2 041 participants were vaccinated in the delayed arm). The median age of consenting CCCs was 35 years old. The final primary analysis included 2 108 participants (51 clusters) vaccinated in the immediate arm and 1 429 participants (46 clusters) eligible and consented on Day 0 in the delayed arm.

The final primary analysis was to assess efficacy against laboratory confirmed EVD by comparing incidence of cases occurring 10 to 31 days post-randomisation for those vaccinated in the immediate vaccination rings versus incidence of cases for participants who consented on Day 0 in the delayed vaccination rings. Vaccine efficacy was 100% (unadjusted 95% CI: 63.5% to 100%; 95% CI adjusted for multiplicity: 14.4% to 100%) (0 cases in the immediate arm; 10 cases in 4 rings in the delayed arm). Randomisation was stopped after an interim analysis with a p=0.0036 that did not meet the pre-specified alpha level of 0.0027. Of the 10 cases, 7 were in contacts, and 3 in contacts-of-contacts. Uncertainties remain as to the level, duration and type of protection given the methodological limitations and the exceptional circumstances experienced during the study.

<u>Clinical immunogenicity</u>

No immune correlates of protection have been defined.

Protocol 009, named Partnership for Research on Ebola Vaccines in Liberia (PREVAIL) was a Phase 2 randomised, double-blind, placebo-controlled study which evaluated the safety and immunogenicity of Ebola vaccine candidates including Ervebo. This study compared Ervebo to normal saline placebo in 1 000 adults ≥18 years of age in Liberia.

Protocol 011, named Sierra Leone Study to Introduce a Vaccine against Ebola (STRIVE) was a Phase 2/3 randomised open-label study which evaluated safety and immunogenicity of Ervebo in adults ≥18 years of age working in healthcare facilities or on frontline activities related to the Ebola response in Sierra Leone. In this study, 8 673 adult participants were enrolled and 8 651 with valid consents randomised to immediate (within 7 days of enrolment) or deferred (18 to 24 weeks after enrolment) vaccination with Ervebo. An immunogenicity sub-study included 508 participants who were vaccinated and provided samples for the assessment of immunogenicity.

Protocol 012 was a Phase 3 randomised, double-blind, placebo-controlled study which evaluated the safety and immunogenicity of three consistency lots and a high dose lot (approximately five times higher than the dose in consistency lots and dose used in other Phase 2/3 studies) of Ervebo compared to normal saline placebo. A total of 1 197 healthy participants 18 to 65 years of age were enrolled in the US, Canada, and Spain.

Protocol 015, named African-Canadian Study of HIV-Infected Adults and a Vaccine for Ebola (ACHIV-Ebola), was a Phase 2, randomised, multi-centre, double-blind, placebo-controlled trial which evaluated the safety and immunogenicity of Ervebo in adult and adolescent participants living with HIV on antiretroviral therapy with an undetectable viral load and a CD4+ T-lymphocyte count greater than or equal to 200 cells/mm^3 who received: a single dose of Ervebo (n=161) or placebo (n=39), or two doses of Ervebo (n=40) or placebo (n=10) administered 56 days apart. In this trial, 65 adolescents 13 to 17 years of age and 186 adults 18 years of age and older were enrolled in Canada, Burkina Faso, and Senegal.

Protocol 016, named Partnership for Research on Ebola VACcination (PREVAC), was a Phase 2 randomised, double-blind, placebo-controlled study which evaluated the safety and immunogenicity of Ervebo in participants who received: a single dose of Ervebo and normal saline placebo administered 56 days apart, two doses of Ervebo administered 56 days apart, or two doses of normal saline placebo. In this study, 998 children and adolescents 1 to 17 years of age and 1 004 adults 18 years of age and older were enrolled in Guinea, Liberia, Mali and Sierra Leone.

Protocol 018 was a Phase 3 open-label study conducted in Guinea to evaluate the safety and immunogenicity of Ervebo in vaccinated frontline workers 18 years of age and older that was implemented as Part B of the Phase 3 ring vaccination study for Protocol 010. In this study, a total of 2 115 participants were enrolled and 2 016 participants were vaccinated with Ervebo. An immunogenicity sub-study included 1 217 participants who were vaccinated and provided samples for the assessment of immunogenicity.

Immunogenicity data were obtained in Protocol 009 in Liberia, Protocol 011 in Sierra Leone, Protocol 012 in the United States, Canada, and Europe, Protocol 015 in Canada, Burkina Faso, and Senegal, Protocol 016 in Guinea, Liberia, Mali, and Sierra Leone, and Protocol 018 in Guinea. Gamma irradiation of specimens (from regions involved in Ebola outbreaks) was performed to reduce risk of wild-type Ebola virus infection of laboratory workers, but increased pre-vaccination glycoprotein enzyme-linked immunosorbent assay (GP-ELISA) immune responses by approximately 20% and decreased post-vaccination GP-ELISA and plaque reduction neutralisation test (PRNT) immune responses by approximately 20%. Samples from Protocol 012 and Protocol 015 were not gamma irradiated. Absence of gamma irradiation, lower baseline seropositivity and other factors resulted in a higher immune response in Protocol 012.

Clinical immunogenicity in adults 18 years of age and older

Immunogenicity testing has been performed in Protocol 009, Protocol 011, Protocol 012, Protocol 015, Protocol 016 and Protocol 018, and includes the assessment of binding immunoglobulin G (IgG) specific to purified Kikwit ZEBOV GP by validated GP-ELISA as well as validated neutralisation of vaccine virus by a PRNT.

As shown in Tables 2 and 3, the geometric mean titres (GMT) of GP-ELISA and PRNT increased from pre-vaccination to post-vaccination.

Over 93.8% of vaccine recipients from Protocols 009, 011, 012, 016 and 018 met seroresponse criteria defined as a ≥2-fold increase from baseline and ≥200 EU/mL at any time post-vaccination by GP-ELISA and over 80.0% of participants met seroresponse criteria defined as a ≥4-fold increase from baseline at any time post-vaccination by PRNT. Over 80.3% of participants continued to meet the seroresponse criteria for GP-ELISA and over 63.8% of vaccine recipients continued to meet seroresponse criteria for PRNT at 12 months. The clinical relevance of the immunogenicity data is currently not known.

Table 2: Summary of geometric mean titres for the GP-ELISA in adults 18 years of age and older from Protocols 009, 011, 012, 016 and 018 clinical studies

Time point	GMT (n) [95% CI]				
	Protocol 009†	**Protocol 011†**	**Protocol 012‡**	**Protocol 016†**	**Protocol 018†**
Baseline	120.7 (487) [110.8, 131.5]	92.7 (503) [85.3, 100.9]	<36.11 (696) [<36.11, <36.11]	140.2 (379) [129.0, 152.4]	78.3 (1,123) [74.7, 82.0]
Month 1	999.7 (489) [920.1, 1 086.1]	964.3 (443) [878.7, 1 058.3]	1 262.0 (696) [1 168.9, 1 362.6]	1 241.2 (343) [1 116.4, 1 380.0]	1 106.5 (1 023) [1 053.4, 1 162.2]
Month 6	713.8 (485) [661.4, 770.3]	751.8 (383) [690.6, 818.4]	1 113.4 (664) [1 029.5, 1 204.0]	NA	1 008.8 (75) [849.8, 1 197.6]
Month 12§	664.3 (484) [616.5, 715.8]	760.8 (396) [697.6, 829.8]	1 078.4 (327) [960.6, 1 210.7]	1 088.4 (292) [983.5, 1 204.6]	NA

Month 24	766.3 (441) [705.0, 832.9]	NA	920.3 (303) [820.4, 1 032.3]	NA	NA
Month 36	755.7 (434) [691.6, 825.7]	NA	NA	NA	NA
Month 48	835.4 (400) [769.3, 907.2]	NA	NA	NA	NA
Month 60	785.9 (397) [722.3, 855.2]	NA	NA	NA	NA

The Full Analysis Set population was the primary population for the immunogenicity analyses in Protocols 009, 011 and 018 and consists of all vaccinated participants with serology data and had a serum sample collected within an acceptable day range.
The Per-Protocol Immunogenicity Population was the primary population for the immunogenicity analyses in Protocol 012 and includes all participants who were compliant with the protocol, received vaccination, were seronegative at Day 1, and had a serum sample at one or more timepoints collected within an acceptable day range.
The Per-Protocol Immunogenicity Population was the primary population for the immunogenicity analyses in Protocol 016 and includes all vaccinated participants with serology data who were compliant with the protocol and had a serum sample collected within an acceptable day range.
n=Number of participants contributing to the analysis.
CI=Confidence interval; GP-ELISA = Glycoprotein Enzyme-Linked Immunosorbent Assay (EU/mL); GMT=Geometric mean titre
§Protocol 011 from Month 9-12
†Protocols 009, 011, 016 and 018 used gamma irradiation of specimens to reduce risk of wild-type Ebola virus infection of laboratory workers.
‡Combined consistency lots group

Table 3: Summary of geometric mean titres for the PRNT in adults 18 years of age and older from Protocols 009, 011, 012, 016 and 018 clinical studies

Time point	GMT (n) [95% CI]				
	Protocol 009†	**Protocol 011†**	**Protocol 012‡**	**Protocol 016†**	**Protocol 018†**
Baseline	<35 (451) [<35, <35]	<35 (438) [<35, <35]	<35 (696) [<35, <35]	17.5 (92) [16.7, 18.4]	<35 (1,107) [<35, <35]
Month 1	117.1 (490) [106.4, 128.9]	116.0 (437) [105.7, 127.4]	202.1 (696) [187.9, 217.4]	170.1 (98) [144.1, 200.7]	160.0 (1 024) [151.6, 168.9]
Month 6	76.7 (485) [69.8, 84.2]	95.3 (382) [86.3, 105.3]	266.5 (664) [247.4, 287.0]	NA	117.0 (75) [96.0, 142.6]
Month 12§	100.2 (485) [91.3, 110.0]	119.9 (396) [107.9, 133.2]	271.4 (327) [243.4, 302.7]	144.3 (84) [122.2, 170.4]	NA
Month 24	NA	NA	267.6 (302) [239.4, 299.2]	NA	NA

The Full Analysis Set population was the primary population for the immunogenicity analyses in Protocols 009, 011 and 018 and consists of all vaccinated participants with serology data and had a serum sample collected within an acceptable day range.
The Per-Protocol Immunogenicity Population was the primary population for the immunogenicity analyses in Protocol 012 and includes all participants who were compliant with the protocol, received vaccination, were seronegative at Day 1, and had a serum sample at one or more timepoints collected within an acceptable day range.
The Per-Protocol Immunogenicity Population was the primary population for the immunogenicity analyses in Protocol 016 and includes all vaccinated participants with serology data who were compliant with the protocol and had a serum sample collected within an acceptable day range.
n=Number of participants contributing to the analysis.
CI=Confidence interval; GMT=Geometric mean titre; PRNT=Plaque Reduction Neutralisation Test
§Protocol 011 from Month 9-12
†Protocols 009, 011, 016 and 018 used gamma irradiation of specimens to reduce risk of wild-type Ebola virus infection of laboratory workers.
‡Combined consistency lots group

Immunogenicity in adults and adolescents living with HIV
In Protocol 015, adults and adolescents (13–17 years) living with HIV on antiretroviral therapy with a CD4+ T-lymphocyte count ≥200 cells/mm³ exhibited sustained immune responses following a single dose of Ervebo. At any time point post-vaccination, ≥97.5% of participants met seroresponse criteria by both GP-ELISA (≥2-fold increase from baseline and ≥200 EU/mL) and PRNT (≥4-fold increase from baseline).
In adults, GP-ELISA GMTs rose from 32.3 EU/mL at baseline to 1,170.8 at Month 1, 1,724.0 at Month 6 and 1,942.7 at Month 12. PRNT GMTs increased from 17.8 at baseline to 223.8 at Month 1, 237.5 at Month 6 and 338.0 at Month 12.
In adolescents, GP-ELISA GMTs increased from 36.3 EU/mL at baseline to 1,432.8 at Month 1, 2,256.4 at Month 6 and 2,951.3 at Month 12. PRNT GMTs rose from 17.5 at baseline to 281.9 at Month 1, 293.0 at Month 6 and 440.9 at Month 12.

Paediatric population

Clinical immunogenicity in children and adolescents 1 to 17 years of age

Protocol 016
As shown in Tables 4 and 5, the GMTs of GP-ELISA and PRNT increased from pre-vaccination to post-vaccination. In Protocol 016, 95.7% of participants met seroresponse criteria defined as a ≥2-fold increase from baseline and ≥200 EU/mL at any time post-vaccination by GP-ELISA and 95.8% of participants met seroresponse criteria defined as a ≥4-fold increase from baseline at any time post-vaccination by PRNT. At 12 months following vaccination, 93.2% of participants continued to meet the seroresponse criteria for GP-ELISA and 95.3% continued to meet seroresponse criteria for PRNT. Tables 4 and 5 provide a summary of GMTs for the GP-ELISA and for the PRNT, respectively, by age range.

Immune responses after vaccination with Ervebo in children and adolescents were non-inferior to those in adults at 1 month post-vaccination. The clinical relevance of the immunogenicity data is currently not known.

Table 4: Summary of geometric mean titres for the GP-ELISA in children and adolescents 1 to 17 years of age from Protocol 016 clinical study§

Age	Baseline GMT (n) [95% CI]	Month 1 GMT (n) [95% CI]	Month 12 GMT (n) [95% CI]
1 to <3 years	50.2 (43) [40.2, 62.7]	1 192.1 (45) [827.6, 1 717.1]	1 719.3 (45) [1 245.7, 2 373.1]
3 to <12 years	93.3 (180) [80.6, 108.1]	1 845.1 (171) [1 552.1, 2 193.4]	1 368.4 (153) [1 189.3, 1 574.5]
12 to 17 years	140.0 (128) [120.9, 162.2]	2 103.3 (120) [1 772.2, 2 496.4]	1 451.6 (86) [1 188.6, 1 772.8]

The Per-Protocol Immunogenicity Population was the primary population for the immunogenicity analyses in Protocol 016 and includes all vaccinated participants with serology data who were compliant with the protocol and had a serum sample collected within an acceptable day range.
n=Number of participants contributing to the analysis.
CI=Confidence interval; GMT=geometric mean titre; GP-ELISA=glycoprotein enzyme-linked immunosorbent assay (EU/mL).
Protocol 016 used gamma irradiation of specimens to reduce risk of wild-type Ebola virus infection of laboratory workers.
§1-dose group

Table 5: Summary of geometric mean titres for the PRNT in children and adolescents 1 to 17 years of age from Protocol 016 clinical study§

Age	Baseline GMT (n) [95% CI]	Month 1 GMT (n) [95% CI]	Month 12 GMT (n) [95% CI]
1 to <3 years	17.5 (39) [<0, <0]	321.0 (33) [231.1, 445.7]	494.7 (32) [386.5, 633.3]
3 to <12 years	17.9 (134) [16.9, 18.8]	280.4 (114) [241.3, 325.7]	312.7 (88) [271.0, 360.8]
12 to 17 years	17.5 (111) [17.4, 17.6]	273.3 (119) [237.5, 314.6]	251.7 (85) [215.7, 293.7]

The Per-Protocol Immunogenicity Population was the primary population for the immunogenicity analyses in Protocol 016 and includes all vaccinated participants with serology data who were compliant with the protocol and had a serum sample collected within an acceptable day range.
n=Number of participants contributing to the analysis.
CI=Confidence interval; GMT=Geometric mean titre; PRNT=Plaque Reduction Neutralisation Test
Protocol 016 used gamma irradiation of specimens to reduce risk of wild-type Ebola virus infection of laboratory workers.
§1-dose group

Clinical immunogenicity in participants receiving a booster dose

Although an increase in antibody responses was observed in children and adolescents (n=207), and adults (n=222) after a second dose of Ervebo administered on Day 56 (Protocol 015 and Protocol 016), the increase in antibody titres was not maintained above the antibody titres observed after a single dose (n=426 children and adolescents, n=507 adults) at 12 months post-vaccination.

5.2 Pharmacokinetic properties

Not applicable.

5.3 Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of repeated dose toxicity and toxicity to reproduction and development.

When Ervebo was administered to female rats, antibodies against the vaccine virus were detected in foetuses and offspring, likely due to trans-placental transfer during gestation and with the acquisition of maternal antibodies during lactation, respectively (see section 4.6).

Ervebo administered to female rats had no effects on mating performance, fertility, or embryonic/foetal development.

Ervebo administered to female rats had no effects on development or behaviour of the offspring.

Environmental risk assessment (ERA)

The vaccine virus is a Genetically Modified Organism (GMO). An ERA was conducted to determine the potential impact of this vaccine on human health and the environment. Because this vaccine is based on VSV, a known pathogen in livestock (e.g. horses, cattle, pigs), the risk assessment included species that are relevant for the wild type (wt) VSV backbone of this vaccine.

In a biodistribution study conducted in non-human primates, vaccine virus RNA was detected in lymphoid organs up to 112 days post-vaccination. However, infectious virus was detected at Day 1 and persistent infectious virus was not detected at any subsequent timepoints measured (Days 56, 84 and 112).

Based on transient shedding data in adults and children from 1 year of age (n=5 for children from 1 to <3 years of age), the results of a toxicity study in non-human primates, and lack of horizontal transmission in pigs, the overall risk of Ervebo to human health and the environment is considered negligible. However, as a precaution, vaccinees and caregivers should attempt to avoid exposure of livestock to blood and bodily fluids from vaccinees for at least 6 weeks following vaccination to avoid the theoretical risk of spread of the vaccine virus. For young vaccinees, if possible, soiled nappies can be cleaned with appropriate detergents or disinfectants; disposable nappies can be sealed in double plastic bags and disposed of in household waste for at least 6 weeks following vaccination. People who develop vesicular rash after receiving the vaccine should cover the vesicles until they heal. The vaccination site or any vesicles must be covered with an adequate bandage (e.g., adhesive bandage or gauze and tape). This provides a physical barrier to protect against direct contact with vesicle fluid (see section 4.2). The bandage may be removed when there is no visible fluid leakage.Medical waste and other cleaning materials must not come in contact with livestock in order to avoid unintended exposure to livestock.

See sections 4.4 and 6.6 for further information.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Recombinant human serum albumin
Trometamol buffer
Water for injections
Hydrochloric acid (for pH-adjustment)
Sodium hydroxide (for pH-adjustment)

6.2 Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

6.3 Shelf life

5 years

6.4 Special precautions for storage

Store and transport frozen at -80 °C to -60 °C.

After thawing, the vaccine should be used immediately; however, in-use stability data have demonstrated that once thawed, the vaccine can be stored for up to 14 days at 2 °C to 8 °C prior to use. At the end of 14 days, the vaccine should be used or discarded. Upon removal from the freezer, the vaccine should be marked with both the date that it was taken out of the freezer and also a new discard date [in place of the labelled expiry date (EXP)]. Once thawed, the vaccine cannot be re-frozen.

Keep the vial in the outer carton in order to protect from light.

6.5 Nature and contents of container

1 mL solution for 1 dose in a vial (type I glass) with a stopper (chlorobutyl) and a flip-off plastic cap with aluminium seal.

Pack size of 10 vials.

6.6 Special precautions for disposal and other handling

- The vaccine is stored frozen at -80°C to -60°C and should be removed from the freezer and thawed in less than 4 hours until no visible ice is present. Do not thaw the vial in a refrigerator as it is not guaranteed that the vial will thaw in less than 4 hours. The thawed vial should then be gently inverted several times prior to withdrawal with the syringe.
- The vaccine should appear as a colourless to slightly brownish-yellow liquid with no particulates visible. Discard the vaccine if particulates are present.
- Withdraw the entire content of the vaccine from the vial using a sterile needle and syringe.

If feasible, the waste liquid from eye washes should be collected and decontaminated before discarding into the drain.

Any unused vaccine or waste material should be disposed in compliance with the institutional guidelines for genetically modified organisms or biohazardous waste, as appropriate.

If breakage/spillage were to occur, disinfectants such as aldehydes, alcohols and detergents are proven to reduce viral infection potential after only a few minutes.

7. MARKETING AUTHORISATION HOLDER

Merck Sharp & Dohme B.V.

Waarderweg 39
2031 BN Haarlem
The Netherlands

8. MARKETING AUTHORISATION NUMBER(S)

EU/1/19/1392/001

9. DATE OF FIRST AUTHORISATION/RENEWAL OF THE AUTHORISATION

Date of first authorisation: 11 November 2019
Date of latest renewal: 17 September 2025

10. DATE OF REVISION OF THE TEXT

Detailed information on this medicinal product is available on the website of the European Medicines Agency https://www.ema.europa.eu.

ANNEX II

A. MANUFACTURER OF THE BIOLOGICAL ACTIVE SUBSTANCE(S) AND MANUFACTURER RESPONSIBLE FOR BATCH RELEASE

B. CONDITIONS OR RESTRICTIONS REGARDING SUPPLY AND USE

C. OTHER CONDITIONS AND REQUIREMENTS OF THE MARKETING AUTHORISATION

D. CONDITIONS OR RESTRICTIONS WITH REGARD TO THE SAFE AND EFFECTIVE USE OF THE MEDICINAL PRODUCT

A. MANUFACTURER OF THE BIOLOGICAL ACTIVE SUBSTANCE(S) AND MANUFACTURER RESPONSIBLE FOR BATCH RELEASE

Name and address of the manufacturer of the biological active substance(s)

Burgwedel Biotech GmbH
Im Langen Felde 5
30938 Burgwedel
Germany

Name and address of the manufacturer responsible for batch release

Burgwedel Biotech GmbH
Im Langen Felde 5
30938 Burgwedel
Germany

B. CONDITIONS OR RESTRICTIONS REGARDING SUPPLY AND USE

Medicinal product subject to medical prescription.

- **Official batch release**

In accordance with Article 114 of Directive 2001/83/EC, the official batch release will be undertaken by a state laboratory or a laboratory designated for that purpose.

C. OTHER CONDITIONS AND REQUIREMENTS OF THE MARKETING AUTHORISATION

- **Periodic safety update reports (PSURs)**

The requirements for submission of PSURs for this medicinal product are set out in the list of Union reference dates (EURD list) provided for under Article 107c(7) of Directive 2001/83/EC and any subsequent updates published on the European medicines web-portal.

D. CONDITIONS OR RESTRICTIONS WITH REGARD TO THE SAFE AND EFFECTIVE USE OF THE MEDICINAL PRODUCT

- **Risk management plan (RMP)**

The marketing authorisation holder (MAH) shall perform the required pharmacovigilance activities and interventions detailed in the agreed RMP presented in Module 1.8.2 of the marketing authorisation and any agreed subsequent updates of the RMP.

An updated RMP should be submitted:

- At the request of the European Medicines Agency;
- Whenever the risk management system is modified, especially as the result of new information being received that may lead to a significant change to the benefit/risk profile or as the result of an important (pharmacovigilance or risk minimisation) milestone being reached.

ANNEX III

LABELLING AND PACKAGE LEAFLET

A. LABELLING

PARTICULARS TO APPEAR ON THE OUTER PACKAGING

OUTER CARTON - PACK OF 10 VIALS

1. NAME OF THE MEDICINAL PRODUCT

Ervebo solution for injection
Ebola Zaire Vaccine (rVSVΔG-ZEBOV-GP, live)

2. STATEMENT OF ACTIVE SUBSTANCE(S)

One dose (1 mL):
Ebola Zaire Vaccine (rVSVΔG-ZEBOV-GP, live attenuated) ≥72 million pfu

3. LIST OF EXCIPIENTS

Recombinant human serum albumin, trometamol buffer, water for injections, hydrochloric acid, sodium hydroxide. See leaflet for further information.

4. PHARMACEUTICAL FORM AND CONTENTS

Solution for injection
10 vials

5. METHOD AND ROUTE(S) OF ADMINISTRATION

Intramuscular use
Read the package leaflet before use.

6. SPECIAL WARNING THAT THE MEDICINAL PRODUCT MUST BE STORED OUT OF THE SIGHT AND REACH OF CHILDREN

Keep out of the sight and reach of children.

7. OTHER SPECIAL WARNING(S), IF NECESSARY

8. EXPIRY DATE

EXP

9. SPECIAL STORAGE CONDITIONS

Store and transport frozen at -80 °C to -60 °C.
Do not thaw the vial in a refrigerator. Do not refreeze.

Keep the vial in the outer carton in order to protect from light.

10. SPECIAL PRECAUTIONS FOR DISPOSAL OF UNUSED MEDICINAL PRODUCTS OR WASTE MATERIALS DERIVED FROM SUCH MEDICINAL PRODUCTS, IF APPROPRIATE

This product contains genetically modified organisms.
Any unused vaccine or waste material should be disposed of in compliance with the institutional guidelines for genetically modified organisms or biohazardous waste, as appropriate.

11. NAME AND ADDRESS OF THE MARKETING AUTHORISATION HOLDER

Merck Sharp & Dohme B.V.
Waarderweg 39
2031 BN Haarlem
The Netherlands

12. MARKETING AUTHORISATION NUMBER(S)

EU/1/19/1392/001 - pack of 10

13. BATCH NUMBER

Lot

14. GENERAL CLASSIFICATION FOR SUPPLY

15. INSTRUCTIONS ON USE

16. INFORMATION IN BRAILLE

Justification for not including Braille accepted

17. UNIQUE IDENTIFIER – 2D BARCODE

2D barcode carrying the unique identifier included.

18. UNIQUE IDENTIFIER - HUMAN READABLE DATA

PC
SN
NN

MINIMUM PARTICULARS TO APPEAR ON SMALL IMMEDIATE PACKAGING UNITS

VIAL LABEL

1. NAME OF THE MEDICINAL PRODUCT AND ROUTE(S) OF ADMINISTRATION

Ervebo solution for injection
rVSVΔG-ZEBOV-GP, live

2. METHOD OF ADMINISTRATION

IM

3. EXPIRY DATE

EXP

4. BATCH NUMBER<, DONATION AND PRODUCT CODES>

Lot

5. CONTENTS BY WEIGHT, BY VOLUME OR BY UNIT

1 dose – 1 mL

6. OTHER

This product contains GMO.

B. PACKAGE LEAFLET

Package leaflet: Information for the user

Ervebo solution for injection
Ebola Zaire Vaccine (rVSVΔG-ZEBOV-GP, live)

Read all of this leaflet carefully before you or your child is vaccinated because it contains important information for you or your child.

- Keep this leaflet. You may need to read it again.
- If you have any further questions, ask your doctor, pharmacist or nurse.
- If you or your child gets any side effects, talk to your doctor, pharmacist or nurse. This includes any possible side effects not listed in this leaflet. See section 4.

What is in this leaflet

1. What Ervebo is and what it is used for

- Ervebo is a vaccine for people who are 1 year of age and older.
- Ervebo is given to protect people from getting Ebola virus disease caused by the Zaire Ebola virus, which is a type of Ebola virus. This vaccine will not protect against the other types of Ebola virus.
- Because Ervebo does not contain the whole Ebola virus, it cannot give people Ebola virus disease.

Your doctor, pharmacist or nurse may recommend receiving this vaccine in an emergency involving the spread of Ebola virus disease.

What is Ebola?

- Ebola is a serious disease caused by a virus. If people get Ebola, it can kill them. People catch Ebola from people or animals who are infected with Ebola or who died from Ebola.
- People can catch Ebola from blood and body fluids like urine, stools, saliva, vomit, sweat, breast milk, semen and vaginal fluids of people who are infected with Ebola virus.
- People can also catch Ebola from things that have touched the blood or body fluids of a person or animal with Ebola (like clothes or objects in direct contact).
- Ebola is not spread through the air, water or food.

Your doctor, pharmacist or nurse will talk to you and then together you can decide if you or your child should receive this vaccine.

2. What you need to know before you or your child receives Ervebo

Do not receive Ervebo if you:

- are allergic to the active substance, rice, or any of the other ingredients of this vaccine (listed in section 6).

You should not receive Ervebo if any of the above apply to you. If you are not sure, talk to your doctor, pharmacist or nurse.

Warnings and precautions
This vaccine might not protect everyone who receives it and the length of time you are protected from Ebola by Ervebo is not known.

Continue to follow your doctor, pharmacist or nurse's recommendations to protect yourself from Ebola infection after you get this vaccine.

<u>Hand washing:</u>

Washing your hands correctly is the most effective way to prevent the spread of dangerous germs, like Ebola virus. It reduces the number of germs on the hands and so reduces their spread from person to person.

Proper hand washing methods are described below;
- Use soap and water when hands are soiled with dirt, blood, or other body fluids. There is no need to use antimicrobial soaps for washing hands.
- Use alcohol-based hand sanitiser when hands are not dirty. Do not use alcohol-based hand sanitiser when hands are soiled with dirt, blood, or other body fluids.

<u>In an area affected by Ebola:</u>

While in an area affected by Ebola, it is important to avoid the following:
- Contact with blood and body fluids (such as urine, faeces, saliva, sweat, vomit, breast milk, semen, and vaginal fluids).
- Items that may have come in contact with an infected person's blood or body fluids (such as clothes, bedding, needles, and medical equipment).
- Funeral or burial rituals that require handling the body of someone who died from Ebola.
- Contact with bats, apes and monkeys or with blood, fluids and raw meat prepared from these animals (bushmeat) or meat from an unknown source.
- Contact with semen from a man who had Ebola. You should follow safe sex practices until you know the virus is gone from the semen.

<u>In case of rash:</u>

If you get a rash where the skin is broken after receiving Ervebo, cover it until it heals. Put the used plasters and bandages in a sealed container, if possible, and throw them in the waste bin to make sure that people with a weak immune system or animals do not come into contact with the plasters and bandages.

<u>Taking care of children that have received Ervebo</u>:

For at least 6 weeks after children receive this vaccine, it is important that you wash your hands thoroughly after you have been in contact with blood or body fluids of vaccinated children. If possible clean soiled nappies with appropriate detergents/disinfectants or if using disposable nappies, seal them in double plastic bag and dispose of them in the household waste.

<u>Talk to your</u> doctor, pharmacist or nurse <u>before you receive Ervebo if you:</u>

<u>Have had allergic reactions to vaccines or medicines</u>
- If you have ever had an allergic reaction to a vaccine or medicine, talk to your doctor, pharmacist or nurse before you receive this vaccine.

<u>Have a weak immune system</u>
If your immune system is weak (which means your body is less able to fight off diseases), you might not be able to receive Ervebo. You might have a weak immune system if:
- you have HIV infection or AIDS,

- you are taking certain medicines that make your immune system weak such as immunosuppressants or corticosteroids,
- you have cancer or a blood problem that makes your immune system weak,
- a member of your family has a weak immune system.

If you think you might have a weak immune system, ask your doctor, pharmacist or nurse if you should receive this vaccine. If you do get the vaccine and have a weak immune system, the vaccine may not work as well as in people with a normal immune system.

Are in contact with vulnerable individuals
Tell your doctor, pharmacist or nurse if in the 6 weeks after you receive Ervebo you might be in close contact with or in the same household as:
- babies who are less than 1 year old,
- someone who may be pregnant or breast-feeding,
- someone who has a weak immune system.

This is because you could pass on the virus in the vaccine to them through your body fluids.

Plan to donate blood
- Do not donate blood for at least 6 weeks after you receive this vaccine.

Are in contact with farm animals
Make sure your blood or body fluids do not come into close contact with farm animals for at least 6 weeks after you receive this vaccine. This is because of a possibility that you could pass on the virus in the vaccine to the animals.

Have a fever (high temperature)
- If you have a fever (high temperature), you should talk to your doctor, pharmacist or nurse before receiving Ervebo. The vaccination may have to be delayed until your fever is gone.
- A minor infection such as a cold should not be a problem but talk to your doctor, pharmacist or nurse before receiving Ervebo.

Have a bleeding disorder or bruise easily
- Tell your healthcare worker if you have a problem with bleeding or you bruise easily. Ervebo might make you bleed or bruise where the vaccine is injected.

Testing for Ebola after you receive Ervebo
- You may test positive for Ebola virus after you receive Ervebo. This does not mean that you have Ebola. Tell your doctor, pharmacist or nurse that you have received Ervebo. They might need to do another test.

Children younger than 1 year of age
If your child is under 1 year old, talk to your doctor, pharmacist or nurse. There is no recommendation for use of Ervebo in children younger than 1 year of age.

Other medicines and Ervebo
Tell your doctor, pharmacist or nurseif you are taking, have recently taken or might take any other medicines or vaccines.

No studies have looked at how other medicines or vaccines and Ervebo might interact with each other. Use of Ervebo with other vaccines is not recommended.

If you plan to receive blood or blood products
Do not receive this vaccine at the same time that you get blood or blood products. Ervebo might not work as well if you get blood or blood products 3 months before or up to 1 month after vaccination.

Pregnancy and breast-feeding

- If you are pregnant or breast-feeding, think you may be pregnant or are planning to have a baby, ask your doctor, pharmacist or nurse for advice before you receive this vaccine. They will help you decide if you should receive Ervebo.

- Do not become pregnant for 2 months after you receive Ervebo. Women who are able to become pregnant should use an effective method of birth control. It is not known if Ervebo will harm the mother or the unborn baby. It is also not known if it can pass to the baby through breast milk.

- If you might be in close contact with, or in the same household as someone who may be pregnant or breast-feeding during the 6 weeks after you receive Ervebo, tell your doctor, pharmacist or nurse. This is because you could pass the vaccine to them through your body fluids.

Ervebo contains sodium

This medicine contains less than 1 mmol sodium (23 mg) per dose, that is to say essentially 'sodium–free'.

3. How Ervebo is given

Ervebo is given by a doctor, pharmacist or nurse. It is given as a single injection (dose of 1 mL) in the top of the arm or the outside of the thigh.

If you have any further questions on the use of this vaccine, ask your doctor, pharmacist or nurse.

4. Possible side effects

Like all vaccines, Ervebo can cause side effects, although not everybody gets them.

Serious side effects:

Serious side effects are rare. Get medical care right away if you or your child has symptoms of a severe allergic reaction (anaphylactic reaction), which may include:

- wheezing or trouble breathing,
- swelling of the face, lips, tongue, or other parts of the body,
- generalised itching, redness, flushing or itchy bumps on the skin.

Other side effects in adults 18 years and older

Very common (may affect more than 1 in 10 people)

- Headache,
- Stomach pain,
- Eating less than usual
- Joint pain,
- Muscle aches,
- Fever,
- Feeling tired or sleepy,
- Chills,
- Pain, swelling, or redness at the injection site.

Common (may affect up to 1 in 10 people)

- Feeling dizzy,
- Nausea,
- Mouth sores
- Skin rash,
- Joint pain, swelling or stiffness,

- Itching at the injection site,
- Excessive sweating.

Certain white blood cell counts can decrease below normal after vaccination but this decrease has not resulted in illness and the counts return to normal.

Most side effects go away within a few days. Joint pain and swelling may last for weeks or months in some people. In some people joint pain and swelling may come back after initially going away.

Side effects in children and adolescents 1 to 17 years of age

Very common (may affect more than 1 in 10 people)
- Headache,
- Stomach pain,
- Eating less than usual,
- Muscle aches,
- Fever,
- Feeling tired or sleepy,
- Chills,
- Pain where your child got this vaccine.

Common (may affect up to 1 in 10 people)
- Feeling dizzy
- Nausea
- Mouth sores
- Joint pain,
- Crying
- Swelling or itching where your child got this vaccine
- Excessive sweating.

Uncommon (may affect up to 1 in 100 people)
- Redness where your child got this vaccine.

Tell your doctor, pharmacist or nurse if you or your child gets any of the side effects listed above.

Reporting of side effects
If you get any side effects, talk to your doctor, pharmacist or nurse. This includes any possible side effects not listed in this leaflet. You can also report side effects directly via the national reporting system listed in Appendix V. By reporting side effects you can help provide more information on the safety of this medicine.

5. How to store Ervebo

- Keep this medicine out of the sight and reach of children.
- Do not use this medicine after the expiry date which is stated on the vial label and the outer carton after 'EXP'. The expiry date refers to the last day of that month.
- Store and transport frozen at -80 °C to -60 °C.
- After thawing, the vaccine should be used immediately. However, once thawed, the vaccine can be stored for up to 14 days at 2 °C to 8 °C before use. Discard the vaccine if it is not used by the end of 14 days. Once thawed, the vaccine cannot be re-frozen.
- Upon removal from the freezer, the product should be marked with both the date that it was taken out of the freezer and also a new discard date (in place of the labelled expiry date).
- Keep the vial in the outer carton in order to protect from light.
- Do not use this vaccine if you notice particles in the liquid.

- Do not throw away any medicines via wastewater or household waste. Ask your doctor, pharmacist or nurse how to throw away medicines you no longer use. These measures will help protect the environment.

6. Contents of the pack and other information

What Ervebo contains
The active substance is a living Vesicular Stomatitis Virus. The surface protein of the virus has been replaced with that of Zaire Ebola Virus (rVSVΔG-ZEBOV-GP).

One dose (1 mL) contains:

Ebola Zaire Vaccine (rVSVΔG-ZEBOV-GP[1,2] live, attenuated) ≥72 million pfu[3]
[1]Recombinant Vesicular Stomatitis Virus (rVSV) strain Indiana with a deletion of the VSV envelope glycoprotein (G) replaced with the Zaire Ebola Virus (ZEBOV) Kikwit 1995 strain surface glycoprotein (GP)
[2]Produced in Vero cells
[3]pfu=plaque-forming units

This product contains genetically modified organisms (GMOs).
This vaccine contains a trace amount of rice protein.

The other excipients are: recombinant human serum albumin, trometamol buffer, water for injections, hydrochloric acid, sodium hydroxide (see section 2).

What Ervebo looks like and contents of the pack
- Ervebo is a solution for injection for 1 dose (1 mL) in a vial (type I glass) with a stopper (chlorobutyl) and a flip-off plastic cap with aluminium seal.
- Ervebo is a colourless to slightly brownish-yellow liquid.
- Ervebo is available in a pack of 10 vials.

For any information about this medicine, please contact the Marketing Authorisation Holder

Marketing Authorisation Holder
Merck Sharp & Dohme B.V.
Waarderweg 39
2031 BN Haarlem
The Netherlands

Manufacturer
Burgwedel Biotech GmbH
Im Langen Felde 5
30938 Burgwedel
Germany

This leaflet was last revised in

Other sources of information

Detailed information on this medicine is available on the European Medicines Agency website: https://www.ema.europa.eu.

This leaflet is available in all EU/EEA languages on the European Medicines Agency website.

The following information is intended for healthcare professionals only:

Standard precautions when caring for patients with known or suspected Ebola disease

Vaccination with Ervebo does not eliminate the necessity of standard precautions when caring for patients with known or suspected Ebola disease. **All healthcare workers, and other ancillary providers who have been vaccinated, should not alter their practices with regard to safe injection, hygiene, and personal protective equipment (PPE) after vaccination.**

Standard precautions, as outlined by WHO, include the following:

- Basic hand hygiene
- Respiratory hygiene
- Use of PPE (to block splashes or other contact with infected materials)
- Safe injection practices
- Safe burial practices

Healthcare workers caring for patients with suspected or confirmed Ebola virus should apply extra infection control measures to prevent contact with the patient's blood and body fluids and contaminated surfaces or materials such as clothing and bedding. When in close contact (within 1 metre) of patients with Ebola Virus Disease, healthcare workers should wear face protection (a face shield or a medical mask and goggles), a clean, non-sterile long-sleeved gown, and gloves (sterile gloves for some procedures).

Laboratory workers are also at risk. Samples taken from humans and animals for investigation of Ebola infection should be handled by trained staff and processed in suitably equipped laboratories.

Vaccine administrators should counsel vaccinees to continue to protect themselves with the following measures:

- Hand washing
- Avoid contact with blood and body fluids
- Safe burial practices
- Safe sex
- Avoid contact with bats and non-human primates or blood, fluids and raw meat prepared from these animals (bushmeat) or meat from an unknown source.

Instructions on the handling of the vaccine before administration

- Ervebo is stored frozen at -80 °C to -60 °C and should be removed from the freezer and thawed in less than 4 hours until no visible ice is present. Do not thaw the vial in a refrigerator as it is not guaranteed that the vial will thaw in less than 4 hours. The thawed vial should then be gently inverted several times prior to withdrawal with the syringe.
- After thawing, Ervebo should be used immediately; however, in-use stability data have demonstrated that once thawed, the vaccine can be stored for up to 14 days at 2 °C to 8 °C prior to use. At the end of 14 days, the vaccine should be used or discarded. Upon removal from the freezer, the product should be marked with both the date that it was taken out of the freezer and also a new discard date [in place of the labelled expiry date (EXP)]. Once thawed, the vaccine cannot be re-frozen.
- Ervebo is a colourless to slightly brownish-yellow liquid. Discard the vaccine if particulates are present.
- Ervebo should be administered intramuscularly. Do not inject the vaccine intravascularly. No data are available for administration via the subcutaneous or intradermal routes.
- Ervebo should not be mixed in the same syringe with any other vaccines or medicinal products.
- Withdraw the entire content of Ervebo from the vial using a sterile needle and syringe. The preferred injection site is the deltoid area of the non-dominant arm or in the higher anterolateral area of the thigh. Cover the injection site with gauze or bandage (e.g. any adhesive bandage or

gauze and tape) that provides a physical barrier to protect against direct contact with vesicle fluid. The bandage may be removed when there is no visible fluid leakage.

- Any unused vaccine or waste material should be disposed of in compliance with the institutional guidelines for genetically modified organisms or biohazardous waste, as appropriate. If breakage/spillage were to occur, disinfectants such as aldehydes, alcohols and detergents are proven to reduce viral infection potential after only a few minutes. If feasible, the waste liquid from eye washes should be collected and decontaminated before discarding into the drain.